

ProximityG: A Python package for proximity and biological graphs in the plane

Héctor Maravillo^{*1}, Diego Villareal², and Heriberto Espino²

¹Universidad Autónoma de la Ciudad de México, Mexico

²Universidad de las Américas Puebla, México

Tags

Python; proximity graphs; biological graphs;

Summary

γ -neighborhood graph [8]

α -shapes [3]

β -skeletons [6]

Unit disk graph [1]

RNG [7]

Stepping Stone Graph [5]

Statement of need

Asymptotic analysis of proximity graphs [2]

Probabilistic properties of the k -nearest-neighbors graph [4]

^{*}Corresponding author.

Package overview

Usage example

Acknowledgements

References

References

- [1] Brent N Clark, Charles J Colbourn, and David S Johnson. Unit disk graphs. *Discrete Mathematics*, 86(1–3):165–177, 1990.
- [2] Luc Devroye. The expected size of some graphs in computational geometry. *Computers & Mathematics with Applications*, 15(1):53–64, 1988.
- [3] Herbert Edelsbrunner, David Kirkpatrick, and Raimund Seidel. On the shape of a set of points in the plane. *IEEE Transactions on Information Theory*, 29(4):551–559, 1983.
- [4] David Eppstein, Michael S Paterson, and F Frances Yao. On nearest-neighbor graphs. *Discrete & Computational Geometry*, 17:263–282, 1997.
- [5] Sameera Kannangara, Egemen Tanin, Aaron Harwood, and Shanika Karunasekera. Stepping stone graph: a graph for finding movement corridors using sparse trajectories. *ACM Transactions on Spatial Algorithms and Systems (TSAS)*, 5(4):1–24, 2019.
- [6] David G Kirkpatrick and John D Radke. A framework for computational morphology. In Godfried Toussaint, editor, *Computational Geometry*, pages 217–248. 1983.
- [7] Godfried Toussaint. The relative neighbourhood graph of a finite planar set. *Pattern Recognition*, 12(4):261–268, 1980.
- [8] Remco C Veltkamp. The γ -neighborhood graph. *Computational Geometry*, 1(4):227–246, 1992.